

AI Harness Audit

ia-marketplace — skills conventions audit (scoped: dimension 2 of 8)

PREPARED BY

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DATE

2026-07-13

METHOD

Framework v1.0 — scoped to dimension 2 (Skills), mechanical conventions check

ASSISTED BY

Claude Code (Fable 5)

VERSION

1.0

CLASSIFICATION

Confidential — internal

Document Control

Target	ia-marketplace repo — Claude Code plugin marketplace (9 plugins, 122 skills)
Stack	Markdown skills + JSON manifests (Claude Code plugin format)
Harness scope	Scoped audit: dimension 2 (Skills) conventions only. Contract, guides, sensors, enforcement, memory, improvement loop, and tooling dimensions were not assessed.
Prepared by	Daniel Heler Pohlmann
Engagement date	2026-07-13
Framework	v1.0 (ai-harness-audit @ 257b2e0) — the method version this audit ran against
Method	Mechanical conventions scan (script: assets/audit_skills.py) + manual verification of every flagged item
Base refs	ia-marketplace @ 257b2e0 (branch chore/skills-conventions-audit)
Prior audit	none — first audit of this target (the 2026-06-24 pim audit assessed a consumer of this marketplace)
Report version	v1.0 — Final

1. Executive Summary

The ia-marketplace repository distributes 122 Claude Code skills across 9 domain plugins (dev 41 · specs 21 · stack 19 · security 16 · legal 16 · QA 3 · tools 3 · custom-agent 2 · workflows 1). This engagement audited every skill against the repo’s own documented conventions (CLAUDE.md): naming, frontmatter shape, description discipline, flat directory anatomy, cross-link integrity, and manifest/doc count synchronization.

Verdict. The skill catalog is in strong shape: all 122 skills pass naming (name = directory, kebab-case, ≤ 64 chars), minimal-frontmatter, and flat-anatomy checks, and every plugin.json / marketplace.json skill count is accurate. The audit found **one critical convention violation** (a description over the 1024-char frontmatter limit), **one documentation drift** (QA plugin counts stale in 4 places), and a handful of low-severity hygiene items. All High/Medium findings were **remediated during the engagement** and re-verified by re-running the scan (0 critical, 0 open warnings that are not documented exceptions). The structural gap — convention compliance checked by hand rather than by a versioned sensor — was also closed in-engagement: the repo’s existing scripts/verify_marketplace.py was extended to cover the missed classes (count sync, stray files, cross-links) and is now gated in CI (F5).

Skills dimension: ✓ ok · **Maturity L3 — Defined (target L4).** Conventions are documented, uniformly applied, and now machine-enforced; L4 pends a CI track record on real PRs. **Assurance opinion (scoped): Satisfactory** — post-remediation; as-found state was Satisfactory with one critical exception, fixed same-day.

Dimension	Score	Maturity	One-line
2. Skills (conventions)	✓ ok	L3 — Defined	122/122 pass structural checks; conventions documented; enforcement still manual
1, 3–8 (all others)	—	—	out of scope for this engagement

Top priorities. (1) ~~Wire the conventions sensor to CI~~ — done in-engagement (F5): verifier extended + `.github/workflows/verify-marketplace.yml`. (2) Bring the two `custom-agent` descriptions into the `USE FOR / DO NOT USE FOR` structure. (3) Progressively split the 13 oversized `SKILL.md` files into `references/`.

2. Methodology & Scope

Frame. Agent = Model + Harness. This marketplace repo *is* harness — it ships the skills that steer consumer agents (feedforward). A defect in a skill’s frontmatter is a defect in every consumer’s harness.

Scope. Scoped engagement: dimension 2 (Skills) of the 8-dimension framework, **conventions only**. The checks come directly from the target’s own contract (CLAUDE.md “Naming & invocation” and “Adding a skill”) plus the framework checklist §2:

1. Plugin declared in `marketplace.json` and carries `.claude-plugin/plugin.json`.
2. Flat anatomy: each skill dir holds exactly one `SKILL.md` (one level deep, no nesting), plus only `references/` and `scripts/`.
3. Frontmatter keys $\subseteq \{ \text{name}, \text{description}, \text{allowed-tools} \}$.
4. `name` = directory name, `^[a-z0-9-]+$`, ≤ 64 chars.
5. `description` present, single-line, ≤ 1024 chars, `USE FOR / DO NOT USE FOR` structure.
6. `use <flat-name>` cross-links resolve to an existing skill.
7. “(N skills)” claims in `plugin.json` / `marketplace.json` and the CLAUDE.md / README.md tables match the real directory counts.
8. `SKILL.md` body $\leq \sim 500$ lines (progressive-disclosure guideline; overflow $\rightarrow$ `references/`).

Method. A Python scanner (`assets/audit_skills.py`) walked all 9 plugins and emitted severity-ranked findings (CRIT/WARN/INFO); every flagged item was then verified by hand against the file before being reported (two WARN classes were reclassified on inspection — see F4). The scan was re-run after remediation to confirm closure.

Limitations. Not assessed: skill *content quality* (whether the guidance in each SKILL.md is correct), skill evals, trigger/description effectiveness in live routing, the other 7 harness dimensions, and delivery pinning at consumers (the `pim` audit covers the consumer side). Point-in-time: results bind to commit `257b2e0` plus the same-day remediation commits.

3. Architecture Overview

ia-marketplace/	(reproducibility boundary = the repo itself)
├─ .claude-plugin/marketplace.json	← 9 plugin declarations (counts audited)
├─ <plugin>/	← dev · stack · security · legal · tools · QA ·
│ └─ .claude-plugin/plugin.json	← workflows · specs · custom-agent
│ └─ skills/<skill>/SKILL.md	← 122 skills, flat, + optional references/ scripts/
├─ CLAUDE.md · README.md	← conventions contract + human docs (counts audited)
└─ ai-harness-audit/	← internal tooling (not a plugin; hosts this report)

Consumers register the marketplace as a **local-directory** source and enable plugins per project; skill names are auto-namespaced by plugin at invocation time (`custom-agent:graphify`). Nothing in a skill directory is executed at install time — the audit surface is purely declarative files.

4. Diagnosis

The central tension is **manual convention enforcement in a repo whose whole purpose is conventions**. CLAUDE.md states precise, checkable rules (name = dir, ≤ 1024-char descriptions, count-carrying manifests), and 122 skills follow them almost perfectly — evidence of disciplined authoring. But nothing *guards* the rules: the one critical violation (F1) and the count drift (F2) both entered on normal feature branches and sat undetected until this audit. The framework's own checklist (§2, "the harness guards itself") flags exactly this: a plugin repo should carry its own skill linting. The scanner written for this audit is the seed of that sensor; committing it and gating CI on it converts a recurring manual audit into a standing control (L3 → L4).

5. Item-by-Item Review

5.2 Skills · ✓ ok · Maturity L3

Maturity rationale. Conventions documented in CLAUDE.md and uniformly applied across 122 skills (L3 — Defined); compliance is verified manually per-engagement, with no versioned lint/CI gate (blocks L4 — Quantitatively Managed).

Criteria. CLAUDE.md naming & anatomy rules; framework checklist §2 (when-to-use descriptions, progressive disclosure, plugin-delivered skills in scope, self-guarding sensors).

Condition (evidence). Scanner output over 9 plugins / 122 skills @ 257b2e0 : initial run 27 findings (1 CRIT · 13 WARN · 13 INFO); post-remediation run 23 findings (0 CRIT · 10 WARN — all documented exceptions or scanner false positives · 13 INFO). Full scanner source in assets/audit_skills.py .

Strengths.

- ✓ 122/122 skills: frontmatter name = directory name, kebab-case, ≤ 64 chars (scanner checks B2, zero hits).
- ✓ 122/122 skills: frontmatter restricted to name / description (/ allowed-tools) — no dropped keys (license , references) reappearing.
- ✓ 122/122 skill dirs flat — no nested SKILL.md , no stray AGENTS.md / README.md / metadata.json (one stray dotfile: F3, removed).
- ✓ All 9 plugin.json and marketplace.json "(N skills)" claims match real counts (post F2 fix, docs tables too).
- ✓ 120/122 descriptions carry the USE FOR / DO NOT USE FOR routing structure with valid sibling cross-links.

Findings (worst-first).

- F1 HIGH** — specs-documentation-userguide description exceeded the 1024-char frontmatter limit (1032 chars). ✓ fixed during engagement
 - Cause:* description grew organically (doc-it lineage sentence added) with no length check at authoring time.
 - Consequence:* violates the documented hard limit; risks truncation or rejection by consumers that enforce the schema, silently degrading skill routing.
 - Evidence:* specs/skills/specs-documentation-userguide/SKILL.md frontmatter, scanner check B3 (1032 > 1024). Remediated to 944 chars; re-scan clean.

- **F2 · Medium** — QA plugin skill count stale (says 2, real 3) in 4 documentation locations.
✓ **fixed during engagement**
 - *Cause*: `e2e-functional-testing` was added with manifests updated but the CLAUDE.md/README.md tables and trees missed (step 5 of “Adding a skill” partially skipped).
 - *Consequence*: the contract file (CLAUDE.md) misinforms both humans and agents about plugin contents; erodes trust in the counts everywhere else.
 - *Evidence*: CLAUDE.md structure tree + plugin table; README.md plugin table + structure tree (4 lines). All four updated; re-scan clean.
- **F3 Low** — stray `.graphify_version` file inside `custom-agent/skills/graphify/`.
✓ **fixed during engagement**
 - *Cause*: artifact of the skill’s original user-level installer, carried along when the skill was moved into the plugin.
 - *Consequence*: violates the SKILL.md + `references/` + `scripts/` anatomy; sets a precedent for installer droppings in skill dirs.
 - *Evidence*: scanner check A3; `grep graphify_version` over the skill = 0 references (safe to delete). Removed.
- **F4 Low** — 6 skills cross-link skills from **external marketplaces** (`dotnet-test:code-testing-generator` , `dotnet-data:optimizing-ef-core-queries` , `vue-best-practices` , `vue-testing-best-practices` , `interface-design`), unresolvable within this repo. ✓ **documented during engagement**
 - *Cause*: CLAUDE.md’s cross-link convention only contemplated siblings; authors legitimately routed to external plugins with no documented rule for it.
 - *Consequence*: in consumers without those external plugins, the `DO NOT USE FOR` hand-off points nowhere — mild routing degradation, no failure.
 - *Evidence*: scanner check B5 over `QA/bdd-gherkin` , `QA/e2e-functional-testing` , `QA/playwright-cli` , `dev/dev-backend-data-modeling` , `stack/stack-i18n` , `legal/legal-accessibility` . (7th hit `use platform-specific` in `legal-content-moderation` is prose — scanner false positive.) Exception now documented in CLAUDE.md “Naming & invocation”.
- **F5 · Medium** — the conventions sensor was incomplete and ungated: `scripts/verify_marketplace.py` existed (naming, frontmatter, description limits) but no CI ran it, and it did not check count claims, skill-dir stray files, or cross-link resolution — exactly the classes F1 slipped past (unrun) and F2/F3/F4 slipped past (uncovered). ✓ **fixed during engagement**
 - *Cause*: the linter was written for manual invocation and never wired to CI; later conventions (count claims in docs, cross-link discipline) were documented in CLAUDE.md but never encoded.
 - *Consequence*: F1/F2-class regressions could re-enter on any branch and persist until the next manual audit; capped maturity at L3.
 - *Evidence*: pre-engagement `scripts/verify_marketplace.py` (no count/stray/cross-link checks); no `.github/workflows/` . Remediated: verifier extended with count-claim sync (plugin.json, marketplace.json, CLAUDE.md, README.md tables+trees), skill-dir anatomy (stray files/dirs), and cross-link resolution with a documented external allowlist; negative-tested (planted stray file → exit 1); gated by `.github/workflows/verify-marketplace.yml` on push/PR; authoring step added to CLAUDE.md “Adding a skill”. Errors fail CI; the F6/F7 items remain warnings so the gate lands green.
- **F6 Low** — 2 `custom-agent` descriptions (`graphify` , `tamandua-agents`) lack the `USE FOR / DO NOT USE FOR` structure. ⚠ **open**

- *Cause*: both skills were imported from external origins with their original free-form descriptions.
- *Consequence*: weaker routing signal than the other 120 skills; no explicit hand-off to sibling/overlapping skills.
- *Evidence*: scanner check B4; frontmatter of both SKILL.md files.
- **F7 low** — 13 SKILL.md files exceed the ~500-line progressive-disclosure guideline. **⚠ open**
 - *Cause*: content-rich skills authored inline before the `references/` split practice settled.
 - *Consequence*: larger always-loaded context per invocation; slower for consumers and against the repo's own anatomy guidance.
 - *Evidence*: scanner line counts — worst: `dev-design-patterns-behavioral` (1331), `specs-diagramming-plantuml` (870), `tamandua-agents` (793), `dev-design-patterns-structural` (763), `graphify` (713), plus 8 more between 510–621.

Corrective actions. (carried into §7)

Finding	Action	Impact	Effort	Owner
F1	Trim description to ≤ 1024 chars — done (944)	H	cheap	maintainer
F2	Sync 4 doc counts to 3 — done	M	cheap	maintainer
F3	Delete stray dotfile — done	L	cheap	maintainer
F4	Document external-marketplace cross-link exception in CLAUDE.md — done	L	cheap	maintainer
F5	Promote <code>assets/audit_skills.py</code> to a repo-level <code>scripts/lint-skills.py</code> + CI gate	H	deeper	maintainer
F6	Rewrite the 2 <code>custom-agent</code> descriptions with USE FOR / DO NOT USE FOR	M	cheap	maintainer
F7	Split the 13 oversized SKILL.md into <code>references/</code> , worst-first	M	deeper	maintainer

6. Open Questions

1. Should external-marketplace cross-links (F4) be namespaced (`dotnet-test:code-testing-generator`) as a rule? Today 2 of the 6 use bare flat names (`vue-best-practices` , `interface-design`), which are indistinguishable from broken sibling links to a linter.
2. Where should the CI gate for F5 run — this repo only, or should consumers (e.g. LinkDaily) also validate the marketplace at plugin-install time?
3. Does the `.graphify_version` removal (F3) affect the graphify skill's upstream update flow, or was the stamp only meaningful for user-level installs? (No in-repo reference found; assumed safe.)

7. Conclusion & Prioritized Action Plan

Impact × Effort.

	Low effort	High effort
High impact	F1 trim description (done)	F5 lint script + CI ← next
Medium impact	F6 rewrite 2 descriptions · F2 sync counts (done)	F7 split oversized SKILL.md
Low impact	F3 stray file · F4 document exception (done)	—

Action register.

ID	Finding	Action	Impact	Effort	Quadrant	Owner	Target
F5	no self-guarding sensor	version scanner as <code>scripts/lint-skills.py</code> ; add CI job failing on CRIT/WARN	H	deeper	Major project	maintainer	next sprint
F6	2 free-form descriptions	restructure <code>graphify</code> + <code>tamandua-agents</code> descriptions	M	cheap	Quick win	maintainer	this week
F7	13 oversized SKILL.md	split worst 5 first (≥ 700 lines), rest opportunistically	M	deeper	Fill-in	maintainer	rolling

Closing assessment. For a 122-skill catalog maintained by hand, the as-found defect rate (1 critical, 1 doc drift, ~5 hygiene items) is low, and everything actionable-cheap was closed inside the engagement. The catalog earns ✓ ok / L3 — Defined with a clear, single step to L4: make the conventions self-enforcing (F5). **Assurance opinion (scoped): Satisfactory.**

Appendix A — Inventory

Plugin	Skills	Audited	Notes
dev	41	41	3 oversized SKILL.md (F7)
specs	21	21	F1 (fixed); 5 oversized (F7)
stack	19	19	1 external cross-link (F4)
security	16	16	clean
legal	16	16	1 external cross-link (F4); 1 scanner false positive
QA	3	3	F2 (fixed); external cross-links (F4)
tools	3	3	1 oversized (F7)
custom-agent	2	2	F3 (fixed); F6; 2 oversized (F7)
workflows	1	1	clean
Total	122	122	

Excluded from scope: ai-harness-audit/ (internal tooling, not a plugin); consumer-side settings (extraKnownMarketplaces , enabledPlugins); per-user config.

Appendix B — Scoring Rubric

Per-dimension scale ✓✓ strong / ✓ ok / △ weak / X missing; severity Critical–Low (impact × likelihood); maturity CMMI L1–5; assurance Satisfactory / Partial / Unsatisfactory. Full definitions: framework/scoring-rubric.md .

Appendix C — References

- Target conventions contract: CLAUDE.md ("Naming & invocation", "Adding a skill") @ 257b2e0
- Framework: ai-harness-audit/framework/ v1.0 — METHODOLOGY.md , checklist.md §2, item-template.md , scoring-rubric.md , prioritization.md
- Scanner + raw output: assets/audit_skills.py
- Claude Code plugin/skill format: Anthropic Claude Code docs (plugins, skills, marketplaces)